

## Abstract

This project set out to model the 2-D the dynamics of the launch and recovery of the NATO Submarine Rescue System (NSRS) when subject to sea induced motion and an inputted winch cable force.

To account for the moving reference frame Lagrangian mechanics was applied to solve the mechanics of the system. The system in question developed from the simplest model of a simple pendulum with a stationary pivot to an extensible rigid spring pendulum with damping and extra winch cable force, all in a moving reference frame subject to vertical displacement and rotation.

Due to their non-linearity the governing equations derived from the system analysis were solved numerically using Matlab's built in suite of ordinary differential equation solvers. The solver used was ode45, which uses a Runge Kutta 4<sup>th</sup> order method, as its approximation was similar to the analytical solution of a linear damped simple pendulum.

The outcome of successfully modelling the launch and recovery process using an appropriate simplified model was achieved. From which solutions to the governing equations for two sets of initial conditions were found. The resultant acceleration of the submarine being launch and recovered could be determined from these results. They were compared for the most complex model analysed with and without an additional winch force. The resultant acceleration was less for the model with the winch force.

Apologies in advance if the equations are not as clear as intended. It was a result of copying and pasting them into the report template as they could not directly inputted due to compatibility issues.

Keywords:

- Maritime launch and recovery
- NATO Submarine Rescue System (NSRS)
- Lagrangian mechanics
- Solving ordinary differential equations

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# 1. Introduction and background

## 1.1. *The system*

The NATO Submarine Rescue System (NSRS) is a joint venture between France, Norway and the UK with the purpose of rescuing trapped sailors from distressed submarines [1]. The system is stored at HMNB Clyde, Faslane and consists of three major components. The Intervention Remotely Operated Vehicle (IROV), the Submarine Rescue vehicle (SRV) and the Portable Launch and Recovery System (PLARS). Other support items make up the system, including a transfer under decompression facility [1].

This project is focused on the launch and recovery of the SRV via the PLARS. The SRV is the actual submersible vehicle responsible for the recovery of crew from distressed submarines. The PLARS is a portal frame structure with a safe working load of 30 tonnes that is responsible for deploying and recovering the SRV [1], figure 1 shows the system in operation.

Figure1, Launching the SRV using the PLARS,

<http://www.subescapetraining.org/Rescue%20Assets.html>



On recovery the SRV is lowered and attached to a transfer under pressure unit. This allows the recovered crew to undergo a full decompression cycle to prevent the onset of the bends. Attaching the two units requires alignment to within  $\pm 10\text{mm}$  [2].

The PLARS consists of heave compensated winches and motion dampers to provide sufficient damping to the oscillation of the SRV suspended below it. In its current form the system can operate up to a sea state 6 [1].

If the damping is insufficient accurate alignment won't be possible, sensitive equipment on board the SRV could be damaged and the rescued sailors will be in an uncomfortable position. It's these limitations that currently restrict the operating condition of the NSRS to a sea state 6 [1].

This project will model the system with an additional winch on the deck of the mother ship to further assist the damping mechanisms of the PLARS in stabilising the motion of the SRV when suspended.

This extra winch is not currently in use and the NSRS operates off multiple mother ships so realistic arbitrary dimensions will be used, listed in section 4.1, to overcome this.

## **1.2.      *Project Approach***

The launch and recovery of the SRV was modelled in 2-D as a pendulum. It is a two-degree of freedom system, this being the displaced angle between the suspending cable and the vertical, and the extension of the cable. The dynamics will be explained by differential equations with known initial conditions. A 2-D model was selected to maintain some simplicity when a moving reference plane is applied to the system. In practical use this is the equivalent to the mother ship only being subject to pitch and heave.

A one-degree of freedom simple 2-D pendulum with a fixed reference point is a well-documented model with an analytical solution. This was the initial model studied. For this model the connected cable was assumed to be an inextensible, massless rigid rod. The equations of motion were derived using both Classical and Lagrangian mechanics to provide faith and experience with the Lagrangian approach.

The derived equations of motion were solved both analytically and numerically. The purpose of this was to provide confidence in the computational model that was set up in Matlab, and to aid in the selection of which ordinary differential equation solving algorithm to use.

With the Matlab model up and running the applied assumptions were reduced. These assumptions were reduced one by one with the equations of motion derived at each step. This allowed the comparison between each set of equations to provide confidence in the set-up of each stage of the model.

The two assumptions that will remain in tact through out the project are that the cable is rigid and that it is massless.

The rigid cable was maintained to prevent the need to apply a finite element method to the cable to look at the curvature in it. This was not taken on as it was determined to be infeasible for the time frame given.

The cable will also be assumed to be massless. This idealised model can be used because the mass of the cable is small compared to the 30 tonnes SRV suspended on it [3]. This assumption also suggests that the tension in the cable is constant [3].

The first assumption removed was the lack of damping. This was done using a Rayleigh's dissipative function. This damping coefficient was approximated as to achieve light damping. It is an arbitrary value that could be replaced by a more realistic value if someone was to investigate the damping characteristics of the launch and recovery system. Although this value is only approximate it was unrealistic to attempt to model the damping achieved within the strands of the cable as they interact in the time frame available.

Upon applying damping the next step was to model the cable as a rigid spring. This meant the system was now a two degree of freedom system. The rigidity prevents displacement of the cable in the perpendicular component of the cable while the spring allows movement in the parallel component.

The next assumption removed was the stationary pivot point. The reference frame was subject to independent vertical and rotational displacement and then the two combined, this represented the mother being subject to pitch and or heave.

These equations of motion were solved using over simplified sea induced motion functions to assess the results of the Matlab solver.

The winch cable itself was then added to the model and portrayed as a force. It was assumed that the magnitude of the force is independent of the angle of the winch cable relative to the horizontal. This makes the problem an optimisation problem.

In reality it is an optimal control problem, as the force in the winch cable is dependent on torque in the winch drum. A control algorithm could be determined so that the winch could be incorporated into a negative feedback loop in which the position of the SRV is used to determine the required force the winch should apply. This problem was not undertaken as it was deemed to be infeasible giving the time constraint of this project.

With these functions up and running the next step was to use more realistic approximations for the motion of the mother ship from sea motion. This was achieved using a Pierson-Moskowitz spectrum. Unfortunately this could not be properly incorporated into the final model due to computational difficulties that are later explained in section 3.3.

Matlab was used, as it is capable of solving the differential equations that will govern the models, providing initial conditions are known. Also it has a suite of ordinary differential equation solvers that can be utilised to simplify setting up the script to provide the solutions. The initial conditions will be dependent on the conditions the NSRS is operating in.

The main ambition of this project was to derive and solve the equations of motion of the launch and recovery of the SRV, when it is out of the water, while subject to sea induced motion by modelling it as a 30 tonne point mass suspended at the end of a rigid spring subject to a force to represent an attached winch cable.

Once this model was achieved the acceleration of the SRV could be determined. The acceleration of the SRV is important to the operators of the NSRS as it is this that will cause unwanted further distress to the submariners, due to sea sickness, has the potential to damage sensitive equipment on board and makes achieving the required alignment difficult.

The setting up of the problems and the derivation of the equations of motion will be displayed in chapter 3. The solving of these equations are described in chapter 4. With solutions shown in chapter 5.

All later mentions of a simple pendulum will refer to a pendulum with a cable modelled as a rigid rod with inextensible length and a spring pendulum refers to pendulum modelled as a rigid extensible rod.

## 2. Literature review

The original search for information was for the analysis of spring pendulums and pendulums in moving reference frames. This search returned many of the constituent parts of this project that could be combined throughout its developed but nothing major that could be used complete.

One source used [4] was a mathematical project on the different models that can be applied to pendulum like systems, the simple and spring pendulum which will be used in this project and the use of differential algebraic equations to solve the taught slack pendulum. This study used moored boats as its example and focused on solely vertical displacements of the reference frame. Lagrangian mechanics was used for the spring pendulum model. This same approach was taken in another mathematical project [5] looking at the dynamics of spring pendulums in particular quasi-periodic behaviour of large amplitudes. However this report was for a stationary pivot point.

A piece on simple pendulums in which the pivot point travels in a circular motion was found [6]. This provided a way to look at the rotation of the mother ship about its centre of gravity. The derived equations of motion were different as in this case the pivot moved around a circle with a constant angular velocity, but provided a starting point to understand.

Maplesoft provided a piece on deriving the equations of motion for a simple pendulum in a moving reference plane with Newtonian mechanics and solving them with their software [7].

The relative complexity of the Newtonian method comparative to Lagrangian and the more literature on the Lagrangian method is what led to the decision to use Lagrangian mechanics.

Once this decision was made the following two books were referred to Classical Mechanics [8] and Lagrangian and Hamiltonian mechanics [9]. These provided the basics on how to apply Lagrangian mechanics to a system by defining the generalised co ordinates and then performing the required differentials.

The idea to use a Rayleigh dissipative function to represent the damping came from [4]. It was successfully applied in this piece and it seemed sensible so was decided to be used.



Further reading defined the Rayleigh's dissipative function as a '*function used to handle the effects of velocity-proportional frictional forces in Lagrangian mechanics*' [10]. A damping force fits the velocity proportional description.

The definition of the potential energy associated with the extension of the spring representing was taken from [4]. As the SRV is suspended off a cable not a rod [11] was referred to see how this would affect the spring constant.

Incorporating the winch force was done by the use of virtual work as described in [8] and demonstrated in [12] where it was used to incorporate friction into the Lagrangian model of a block sliding down an inclined plane.

A Pierson-Moskowitz Spectrum was intended to be used to approximate the sea that the launch and recovery will be operating in. A script to create wave height as a function of time from was provided by M.R.Belmont. The spectrum was read up on from [13].

For help with the use of Matlab, Mastering Matlab 7 [14] was referred to. This provided information on the use of Matlab's suite of ordinary differential equation solvers and performing numerical differentiation on the software. Further sources were used for help with solving ordinary differential equations [15] solves Blasius' solution to boundary layer flow a single degree of freedom problem and [16] solves a system of differential equations with two degrees of freedom. With the benefit of the problem being a spring pendulum with a stationary pivot. Providing a point to compare the solution of the project at that stage to.

There currently is 3-D simulation software NSRS NIREUS '*a modelling, simulation and prediction tool for NSRS SRV Launch and Recovery Operations*' [17]. It lacks the additional winch this project will incorporate [18].

To summarise the models in this project incorporating a winch force were derived from scratch using [12] to derive the terms and incorporate them into the governing equations, [19] provided the necessary vector formulae and [20] for some of the differentiation.

Models involving both translation and rotation of the reference frame are newly derived and were done so by combining the definitions of the position vector of the mass from [4] and [6] as this provided position co ordinates in terms of generalised co ordinates,

which were selected in these sources. The governing equations were then derived following the method explained in [8] and [9].

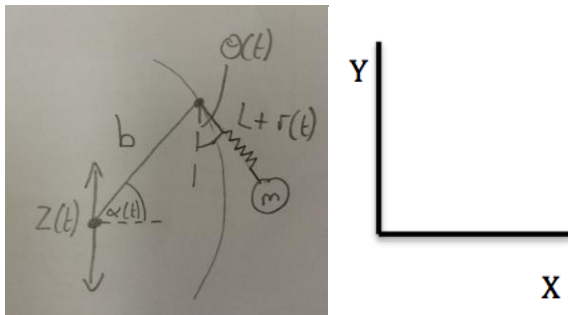
### 3. Methodology and theory

The governing equations were derived following the method described in section 3.1. Governing equations were calculated for all variations of the model in question. The simplest model also provided a reference point to compare Newtonian and Lagrangian results for confidence in the new method. This allowed them to be cross-referenced to provide confidence in the equations. For example the equations of motion for the spring pendulum when the extension is set to zero should equal the equations of motion for the simple pendulum under the same conditions.

As stated in chapter 2 some sources already provided the governing equations for some of the systems of interest of this project. For the models full derivations were still carried out as the reports they were found in did not show full derivations and it would be unwise to just assume they are correct. However for comparison they provide confidence in the resulting equations.

#### 3.1. *Deriving the Governing Equations*

Figure 2, The spring pendulum model with co ordinate axis



From figure 2 the position vectors of any of the models used in this project can be derived by setting any combination of  $b$  and  $\alpha(t)$  representing the pitch function,  $Z(t)$  representing the heave function and  $r(t)$  representing the extension of the spring, to zero. Keeping them all achieves the final model. From this figure the first thing to do is define the kinetic and potential energies.

The position vector of the mass, M, is

$$M = (b\cos\alpha + (L + r)\sin\theta)i + (Z + b\sin\alpha - (L + r)\cos\theta)j \quad \text{eqn1}$$

From this by use of the chain rule the velocity in x and y components can be found.

$$\frac{dx}{dt} = \frac{dx}{d\theta} * \frac{d\theta}{dt} \quad \text{eqn 2}$$

Once the chain rule has been applied to both the x and y components of the position vector the results were then squared and summed. Trigonometric identities were used to simplify the more complicated expressions.

$$|v^2| = (\dot{x}^2 + \dot{y}^2) \quad \text{eqn3}$$

The Kinetic energy, T, can now be calculated:

$$T = \frac{1}{2}m|v^2| \quad \text{eqn4}$$

The potential energy, V, is composed of two parts the gravitational potential energy and the elastic potential energy of the spring. For the simple pendulum models only the gravitational potential energy is needed. For the elastic potential energy the extension needs to be defined. This extension definition is from [4].

When the  $\theta = 0$  and the pendulum is in the equilibrium position the length of the cable is:

$$L = \frac{mg}{k} + L_0 \quad \text{eqn 5}$$

Thus once the SRV is attached to the cable, as it will be during our analysis, the extension in the cable at any time is:

$$\Delta L = \frac{mg}{k} + r \quad \text{eqn6}$$

From this the potential energy is:

$$V = mgy + \left(r + \frac{mg}{k}\right)^2 \quad \text{eqn7}$$

The kinetic and potential energies could then be combined into the Lagrangian

$$L = T - V \quad \text{eqn 8} \quad [8]$$

From there the models without any damping were defined using the Lagrange's equation for a conservative standard system.

$$\frac{d}{dt} \left( \frac{\delta L}{\delta \dot{q}_n} \right) - \frac{\delta L}{\delta q_n} = 0 \quad \text{eqn 9 [8]}$$

Where  $q_n$  represents a generalised co ordinate. In the models for this project the generalised co ordinates are the angle between the cable supporting the mass and the vertical,  $\theta$ , and the extension of the cable,  $r$ .

Damping could then be incorporated into the model using a Rayleigh's dissipative function. This was chosen because it has been successfully applied previously in other projects [4] and for the spring pendulum models can also represent the viscous damping in the cables as they tighten and extend and individual strands interact with one another and the oil they're coated in.

$$F_d = \beta |\dot{v}|^2 \quad \text{eqn 10 [4]}$$

Differentiating this with respect to the generalised co-ordinates allows them to be substituted into equation 11 along with the derivatives calculated for equation 9.

$$\frac{d}{dt} \left( \frac{\delta L}{\delta \dot{q}_n} \right) - \frac{\delta L}{\delta q_n} + \frac{\delta F}{\delta \dot{q}_n} = 0 \quad \text{eqn 11 [8]}$$

The final step is to then add the winch force to system for defining the governing equations. This was achieved using equation 12. Where  $W$  represents the virtual work done by the force to the generalised co ordinate. The derivation of this is shown in section 3.2.

$$\frac{d}{dt} \left( \frac{\delta L}{\delta \dot{q}_n} \right) - \frac{\delta L}{\delta q_n} + \frac{\delta F}{\delta \dot{q}_n} = \frac{\delta W}{\delta q_n} \quad \text{eqn12 [12]}$$

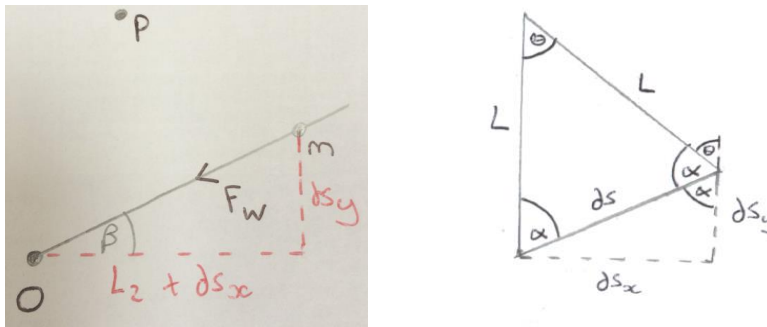
### **3.2. Deriving the effect of the winch force on the system**

The winch cable was represented as a force,  $F_w$ . As previously stated the magnitude of the force will be independent of the angle between the cable and the horizontal. The winch force can only be applied when the mass is swinging away from the ship as this keeps the cable taught. Assuming that theta is positive, as the mass swings anti-clockwise, away from the mother ship, This is equivalent to the force only being applied when  $\theta' > 0$ .

The effect of the winch force was incorporated into the Lagrangian model using virtual work. It was split into two components one for each generalised co-ordinate. This could be done as the virtual work functions are differentiated with respect to generalised co-ordinates, and the individual generalised co-ordinates are independent of one another [8].

### ***Effect on Theta***

Figure 3 determining the angle of the winch force and the horizontal,  $\beta$



The displacement of the mass  $ds$  was defined using the cosine rule. The extension,  $r$  was deemed negligible because running models before the addition of the winch force gave a maximum extension of 26mm, which is minute, compared to the length of the cable 7m. This simplified the derivations and equations that needed to be input into Matlab.

The co-ordinates of the pivot point  $P$ , (3,7), were decided from potential dimensions of the ship with the top of the winch drum, as the origin, at the same height and a distance of  $L_2$  from the centre of mass of the SRV when  $\theta$  is zero. These are arbitrary values as the mother ship is never the same and none have this extra winch. However for this project they are from figure 5 in section 4.1, where the constants are defined.

The translation and or rotation of the reference frame caused by the sea state did not need to be incorporated for these vector definitions. The pitch and heave of the ship is the equivalent to the translation and rotation of the ship reference frame within a larger world reference frame. As the ship reference frame moves the relative vectors within the ship reference frame remain constant.

The Virtual work of the winch force on  $\theta$  is the product of the winch force,  $F_w$ , the perpendicular distance between the line of action of this force and  $P$ ,  $h$ , and the virtual

angular displacement,  $\theta$ . The virtual work will be negative because the winch force acts to decrease theta. Applying the product rule to this gives equation 13:

$$\frac{\delta W}{\delta \theta} = -F_w * \left( h + \left( \frac{dh}{d\theta} * \theta \right) \right) \text{ eqn 13}$$

The perpendicular distance, h was calculated using a formula for determining the perpendicular distance between a point and a line [19]

$$h = \frac{|(3 + (L_z + ds_x)) + (7 * ds_y)|}{\left( (L_z + ds_x)^2 + (ds_y)^2 \right)^{0.5}} \text{ eqn 14}$$

The horizontal and vertical components of the displacement of the mass were defined from figure 3 by using simple trigonometry and using the cosine rule to define the displacement to give:

$$ds_x = (2L^2 - 2L \cos \theta)^{0.5} * \sin \alpha \text{ eqn 15.1}$$

$$ds_y = (2L^2 - 2L \cos \theta)^{0.5} * \cos \alpha \text{ eqn 15.2}$$

$$\text{Where: } \alpha = \frac{(\pi - \theta)}{2}$$

The differentiation of the perpendicular distance was done with the quotient rule. The numerator being a modulus meant that differentiating moduli had to be looked up [20], giving the following relationship:

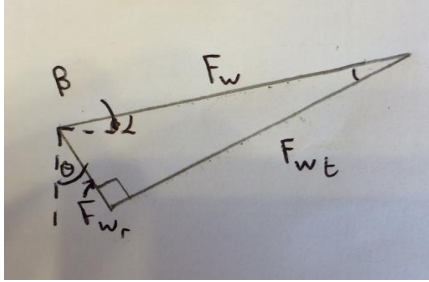
$$g(\theta) = |u| \quad g'(\theta) = \frac{u * u'}{|u|} \text{ eqn 16}$$

The differentiation of u with respect to theta is provided, equation 17, to provide confidence in the differential procedure, as this project was highly dependent upon differentiation including multiple uses of the chain and products rule.

$$\begin{aligned} u'(\theta) = 3 * & \left[ \left( (0.5 + (2L^2 - 2L \cos \theta)^{-0.5}) * (2L \sin \theta) * \left( \sin \frac{\pi - \theta}{2} \right) \right) + \left( ((2L^2 - 2L \cos \theta)^{0.5}) * \right. \right. \\ & \left. \left( -0.5 * \cos \frac{\pi - \theta}{2} \right) \right) \right] \\ & + \\ & 7 * \\ & \left[ \left( (0.5 + (2L^2 - 2L \cos \theta)^{-0.5}) * (2L \sin \theta) * \left( \cos \frac{\pi - \theta}{2} \right) \right) + \right. \\ & \left. \left( ((2L^2 - 2L \cos \theta)^{0.5}) * \left( 0.5 * \sin \frac{\pi - \theta}{2} \right) \right) \right] \text{ eqn 17} \end{aligned}$$

### Effect on extension

Figure 4, Resolving the winch force into radial and tangential components



The principle of virtual work was again applied. Multiplying the radial component of the winch force by a virtual displacement, the extension  $r$ . Differentiating this with respect to the generalised co ordinate  $r$  gives the radial component of the winch force:

$$\frac{\delta W}{\delta r} = F_w \cos \left( \left( \frac{\pi}{2} - \theta \right) + \beta \right) \text{ eqn 18}$$

Where

$$\beta = \tan^{-1} \left( \frac{(2L^2 - 2L \cos \theta)^{0.5} * \left( \cos \frac{\theta - \pi}{2} \right)}{L_2 + \left( (2L^2 - 2L \cos \theta)^{0.5} * \left( \sin \frac{\theta - \pi}{2} \right) \right)} \right) \text{ eqn 19}$$

The expressions calculated from equation 18 and 13 were substituted into equation 12 along side other expressions as described in section 3.2 to give the governing equations to be inputted into Matlab. Equation 20.1 and 20.2 show the governing equations of the final model, spring pendulum subject to pitch and heave and the additional winch force.

$$\begin{aligned} g'' = & \frac{u'b\theta'(\cos \theta - \alpha)}{(L+r)} - \frac{u'b(r-r') \sin(\theta - \alpha)}{(L+r)} - \frac{Z'\alpha' b \sin \alpha}{(L+r)} - \frac{(y+Z'') \sin \theta}{(L+r)} - \frac{2r'\theta'}{(L+r)} \\ & - \frac{b\alpha'(\theta' - \alpha'') \cos(\theta - \alpha)}{(L+r)} - \frac{b\alpha'' \sin(\theta - \alpha)}{(L+r)} - \frac{\beta\theta'}{m} - \frac{\beta Z' \sin(\theta - \alpha)}{m(L+r)} \\ & - \frac{\beta\alpha' b \sin(\theta - \alpha)}{m(L+r)} + \left( \frac{\frac{\delta W}{\delta \theta}}{m(L+r)^2} \right) \text{ eqn 20.1} \end{aligned}$$

$$\begin{aligned} r'' = & Z'' \cos \theta + \theta'^2 (L+r) + u'b\theta' \sin(\theta - \alpha) + u'br \cos(\theta - \alpha) + g \cos \theta - g - \frac{kr}{m} - \frac{\beta r'}{m} \\ & + \frac{\beta Z' \cos \theta}{m} + \left( \frac{\frac{\delta W}{\delta r}}{m} \right) \text{ eqn 20.2} \end{aligned}$$

In the interest of space the winch force effect was represented by the derivatives in equations 20.1 and 20.2. Full derivations and expressions are available.

### **3.3.      *Resultant acceleration***

The resultant acceleration of the SRV in the final models was calculated by differentiating the resultant velocity with respect to time. This was done because Matlab's suite of ordinary differential equation (ode) solvers only solves first order differential equations. So the highest order of the solutions was velocity, which could be combined to find the resultant and then differentiated as opposed to summing accelerations.

### **3.4.      *Modelling the sea***

The initial plan of using a Pierson Moskowitz spectrum to model the sea model could not be incorporated into the final model. The arrays for the solutions for the first and second derivatives of height with respect to time from the Pierson Moskowitz approximation would not fit with the continuous time domain set for defining the equations of motion. The script of incorporating the more accurate sea model into the solution of the final model is available.

As a result of this it was decided to stick with a sinusoidal approximation to model the sea even though the original intention was just to assess the model with this approximation as it developed to the final model. There was a benefit to this however meaning that the pitch function could still be incorporated into the final SRV model. As if the Pierson Moskowitz approximation was used a function for the pitch would be lost as it was unfeasible to attempt to create an approximation of this from the spectrum too with the time frame.

For the sinusoidal sea model it is assumed that when a swell reaches the centre of gravity of the ship the whole ship is raised and when the swell reaches either the stern or bow of the ship it raises that end of the mother ship about the centre of gravity. Which was assumed to be in the centre of the ship, as the mass of the NSRS and its supporting equipment will balance out the weight of the bridge being towards the bow of the ship. A limiting factor of this approximation is that it does not account for the random nature of wave motion like the Pierson Moskowitz spectrum model would have.

The swell of 4m with a period of 10s will be used as this replicates a sea state 5 [21]. If the stern of ship is raised by 4m about the centre of gravity, the angle between a vector



connecting the SRV pivot point to the centre of gravity and the horizontal is 0.304 radians at its maximum.

The pitch function will be 180 degrees out of phase of the heave function. This means that the centre of gravity will have zero displacement when the pitch is at its maximum. This means that for this model the bow of the ship will be in the trough of a wave, the centre of gravity at a zero point and the stern at a peak. Or the stern and bow at the other extremity of the wave.

Although this phase simplification isn't necessary accurate it provides the greatest angle alpha. Creating the maximum angle is good for a worse case scenario because in some circumstances the random factor of ocean waves could lead to this being the case. So it is best to analyse the dynamics of the system in this case.

$$Z(t) = 4\sin 0.63(t) \text{ Heave function, eqn 21}$$

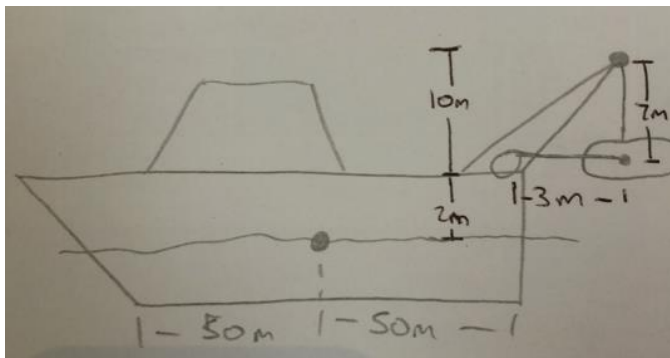
$$\alpha(t) = 4\cos 0.63(t) \text{ Pitch function, eqn 22}$$

## 4. Experimental work/analytical investigation/ design

Having defined the governing equations for the model in question the next step was to define the constants in the equations and to then assess the numerical approximation to the system in question. The constants are arbitrary as there is no exact system to copy but are realistic. From there the Matlab code was developed from solving the governing equation of the simplest model through to the final model, reducing the assumptions as described in section 1.2.

### 4.1. *Defining constants*

Figure 5, Sketch of position of SRV relative to ship for governing equations of this project



The length of the cable when  $\theta=0$  will be 7m, the mass of the SRV is 30 tonnes, the acceleration due to gravity,  $g=9.81 \text{ ms}^{-2}$ . These dimensions were based on a sketch of Northern River, a ship the SRV can operate from, provided by M.R.Belmont. The distance of the SRV behind the mother ship gives the required clearance [18].

### ***Damping coefficient, $\beta$***

In reality this system would be lightly damped. To achieve this the poles of the transfer function must be complex conjugate pairs with negative real parts. This was applied to the equation of motion of the simple damped pendulum in a fixed reference plane. Substituting the constants into the quadratic formula shows that for light damping:

$$71506 \gg \beta$$

From this a damping coefficient of  $30000 \text{ kg s}^{-1}$  was decided upon as it matches the light damping criteria and reaches a steady state response in about 11s, which is reasonable for a system of this scale taking into account the damping equipment present in the PLARS [1]. The units for the damping were derived applying dimensional analysis to the Rayleigh's dissipative function.

### ***Spring constant, $k$***

Because of steel cable's fibrous nature its Young's modulus is less than that of a steel rod of the same diameter, also the effective area is less than the nominal area of the rod [11]. Therefore values for these were taken from [11]

A cable of nominal diameter 38mm was selected, as its ultimate load is 930kN so can take the load of the SRV, 29.4 kN, with a factor of safety of 3.2. For this cable the spring constant is:

$$k = \frac{(140 \times 10^9) + (0.000697)}{7} = 13940000 \text{ Nm}^{-1} \text{ eqn 23 [11]}$$

## ***4.2. Initial conditions***

In some of the models there are four sets of initial conditions. This provides the potential for multiple initial conditions. Therefore it was decided to utilise two potential sets of initial conditions, solution 1 and solution 2, to reduce the workload to a sensible amount. These are also reasonable initial conditions.

#### Solution 1

|                                     |                                     |
|-------------------------------------|-------------------------------------|
| $\theta(0) = 0.133 \text{ radians}$ | $\theta'(0) = 0 \text{ rad s}^{-1}$ |
| $r(0) = 0.000188 \text{ m}$         | $r'(0) = 0 \text{ ms}^{-1}$         |

#### Solution 2

|                                 |                                         |
|---------------------------------|-----------------------------------------|
| $\theta(0) = 0 \text{ radians}$ | $\theta'(0) = 0.779 \text{ rad s}^{-1}$ |
| $r(0) = 0 \text{ m}$            | $r'(0) = 0.0607 \text{ ms}^{-1}$        |

These initial conditions for solution one were derived from figure 5 assuming a factor of safety that the SRV is always more than a metre from the stern of the ship, providing initial angles by use of the cosine rule. The initial extension was then taken by resolving forces to calculate the tension in the cable at an angle of 0.133 radians with a vertical force for the mass of the SRV. Then using the extension definition, equation 6 replacing  $mg$  with the tension, and subtracting  $L$  the initial  $r$  could be found.

For solution two the initial accelerations were calculated so that the kinetic energies of each generalised co ordinate at  $t=0$  were equal to the potential energies at  $t=0$  for solution one. This is a fair assumption because in each circumstance the initial energy should be the same to adhere to conservation of energy.

A third solution, solution 3, was also used where all initial conditions are zero. This provided a basis to compare the steady state response of the solutions with initial conditions.

### 4.3. *Selecting the numerical solver*

The governing equation for a simple damped pendulum in a fixed reference frame was linearised. This was achieved by limiting the range of  $\theta$  this allows for the assumption that for small angles  $\sin\theta \approx \theta$ , for this to be true an initial displacement of 0.1 radians was used.

$$\theta'' + \frac{\beta\theta'}{m} + \frac{g}{L}\theta \quad \text{eqn 24} \quad \text{linearised simple pendulum}$$

For this purpose  $\beta$  was set to 10000 because this takes longer for theta to reach its steady state response. This provides a longer domain of oscillation to assess the numerical solvers

over. With the following initial conditions,  $\theta(0) = 0.1$  and  $\theta'(0) = 0$  solving the equation gives with Laplace transforms gives:

$$\theta(t) = 0.1e^{\frac{-t}{6}}\cos 1.17t + 0.0142e^{\frac{-t}{6}}\sin 1.17t \quad \text{eqn 25}$$

This analytical solution provided a comparison to the results of different numerical solutions. The numerical solver first tried was ode45 ‘an explicit one-step Runge Kutta medium order (4<sup>th</sup>-5<sup>th</sup>-order) solver’ [14] this solver was used first because it is described as typically the first solver to try on a new problem [14].

This numerical solution perfectly mapped the analytical result so there was no need to try any other ordinary differential equation solvers within Matlab, shown in figure 6. With an accurate solver selected the next step was to test it under the other potential set of initial conditions, zero initial displacement and an initial angular velocity,  $\theta(0) = 0$  and  $\theta'(0) = 0.1$ .

For the solution to be successful conservation of energy must be obeyed. With damping present the kinetic energy at  $t=0$  should be greater than the potential energy of the system at  $t = \frac{T}{4}$ . A quarter of a period was selected as it is at this time that the mass will be at its greatest height of swing and therefore greatest potential energy. For this to uphold  $\theta\left(t = \frac{T}{4}\right) < 0.0845$  radians, figure 7 shows this to be true.

Figure 6, numerical solver result overlapped onto analytical

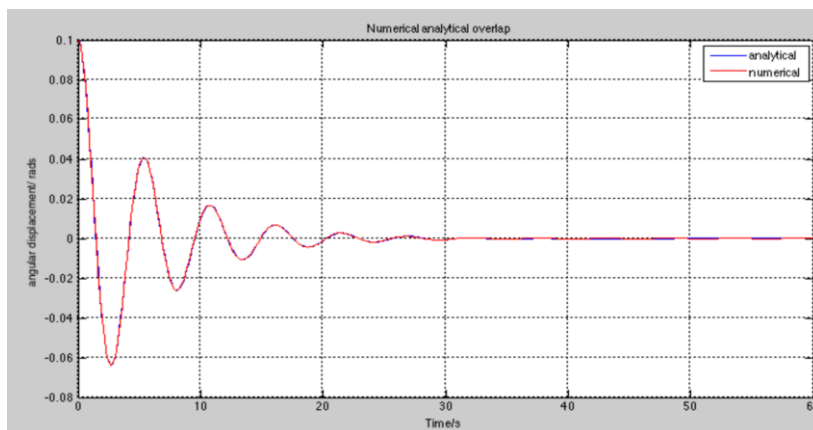
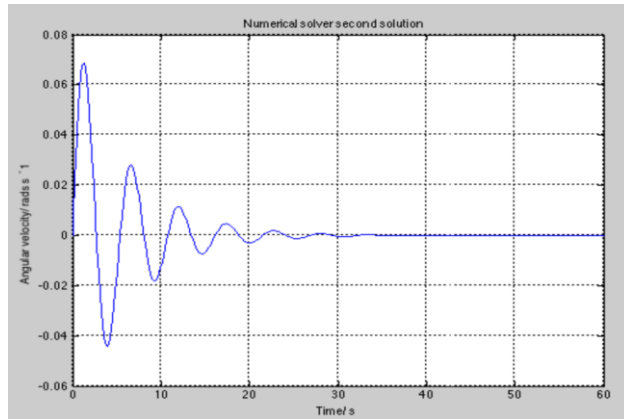


Figure 7, numerical solver with second set of initial conditions



#### 4.4. *Solving the governing equations*

With the constants defined, a numerical solver selected and initial conditions decided upon Matlab could now be utilised to find solutions to the governing equations previously derived.

This was achieved by defining the equations of motion in a Matlab script and saving the script with a suitable name so it could easily be found again. A sample script is provided below figure 8. In the script  $x(1)$  refers to  $\theta$ ,  $x(2)$   $\theta'$ ,  $x(3)$  is the extension  $r$  and  $x(4)$  is  $r'$ .

Figure 8, Script defining governing equation for a spring pendulum in a vertically oscillating reference plane with winch force.

```
function dx=f(t,x)
m=30000;
g=9.81;
L=7;
K=13940000;
B=30000;
Z1=2.52*cos(0.63*(t)); %Z'(t)
Z2=1.59*sin(0.63*(t)); %Z''(t)
%
dx=zeros(4,1);
if (x(2))>0
dx(1)=x(2);
```

$$dx(2) = -((g+Z2)*(\sin(x(1)))/(L+(x(3)))) - ((2*(x(4))*(x(2)))/(L+(x(3)))) - ((B*(x(2)))/m) - ((B*Z1*(\sin(x(1))))/(m*(L+(x(3))))) + \left( \frac{\delta \ddot{r}}{\delta t} / m (L+r)^2 \right);$$

$$dx(3) = x(4);$$

$$dx(4) = ((x(2)^2)*(L+(x(3)))) + (g*(\cos(x(1)))) + (Z2*(\cos(x(1)))) - ((K*(x(3)))/m) - g - ((B*(x(4)))/m) + ((B*Z1*(\cos(x(1))))/m) + \left( \frac{\delta \ddot{r}}{\delta t} / m \right);$$

else

$$dx(1) = x(2);$$

$$dx(2) = -((g+Z2)*(\sin(x(1)))/(L+(x(3)))) - ((2*(x(4))*(x(2)))/(L+(x(3)))) - ((B*(x(2)))/m) - ((B*Z1*(\sin(x(1))))/(m*(L+(x(3)))))$$

$$dx(3) = x(4);$$

$$dx(4) = ((x(2)^2)*(L+(x(3)))) + (g*(\cos(x(1)))) + (Z2*(\cos(x(1)))) - ((K*(x(3)))/m) - g - ((B*(x(4)))/m) + ((B*Z1*(\cos(x(1))))/m);$$

end

The differential equation solver, ode45, built into Matlab could then be applied to the governing equation. This was achieved by using the code in figure 9 in the Matlab command window.

Figure 9, Matlab code to solve governing equations

$$x0 = [\theta(0) \ \theta'(0) \ r(0) \ r'(0)] ;$$

$$t0 = [0 \ 60];$$

$$[t,x] = \text{ode45}(@\text{function\_name}, t0, x0)$$

This sets the solver to solve the equations of motion defined in the .m file, in this case represented by 'function name', over a time domain of t0 with initial conditions x0. For all solutions the time domain was 60 seconds as it is a sensible time domain for the system to reach its steady state response. The initial conditions were set to either solution 1,2 or three as described in section 4.2.

As stated in section 3.2 the winch force is only applied to the SRV when it is swinging away from the mother ship. This was incorporated into the Matlab script using the 'if' and 'else' conditions. In the interest of space the virtual work terms differentiated with respect to their generalised co ordinates are just bolted on to end of the script to represent where their proportion of the code was placed.

The initial solving of differential equations was modified from [15] and then when the system moved onto two degrees of freedom [16] was referred to and modified.

#### **4.5. *Finding the Resultant Acceleration***

Figure 10, Matlab code to calculate resultant acceleration from ode solution

```
>> L=7.+(x(:,3));  
>> V1=L.*(x(:,2));  
>> V1=V1.^2;  
>> V2=(x(:,4)).^2;  
>> V=V1+V2;  
>> V=V.^0.5;  
>> A=diff(V)./diff(t);
```

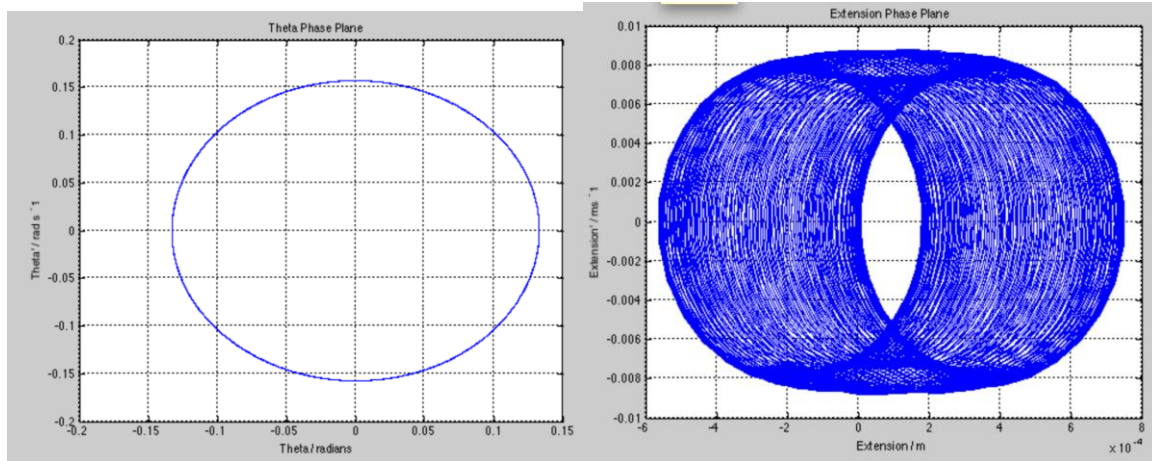
The Matlab function ‘diff’ is the equivalent to using a forward differencing technique [14]. This results in the time array of the acceleration having one less value than the time array. This meant the arrays had to be imported into excel for graph plotting.

## **5. Presentation of experimental or analytical results/descriptions of final constructed product**

### **5.1. *Undamped Spring Pendulum, stationary reference frame***

This was the first two-degree of freedom model and was a sensible one to assess the Matlab code and solver. One of the sources for developing the script [16] solved this same model and provided phase plane diagrams for both the extension and theta. This was the reference point for analysing the results for this model. For further confidence in the scripts developed the damped model was used with  $\zeta$  set to zero, as this should replicate the undamped model.

Figure 11, phase plane diagrams for theta and extension



The solver performed correctly with the phase planes mapping the expected shapes. This provided confidence in the Matlab script to be developed further in line with the reduction of assumptions discussed in section 1.2

## 5.2. *Damped Spring Pendulum, Mother Ship subject to Heave*

For a reference to assess the use of the winch force in section 5.3 and the addition of a pitch function to the model in 5.4 the solutions for the damped spring pendulum in a vertically moving reference frame need to first be analysed. This model was chosen a reference because the solutions are known to be correct on comparison to [4] where the assumptions were used.

Figure 12, Solution of damped spring pendulum, subject to heave for initial conditions 1

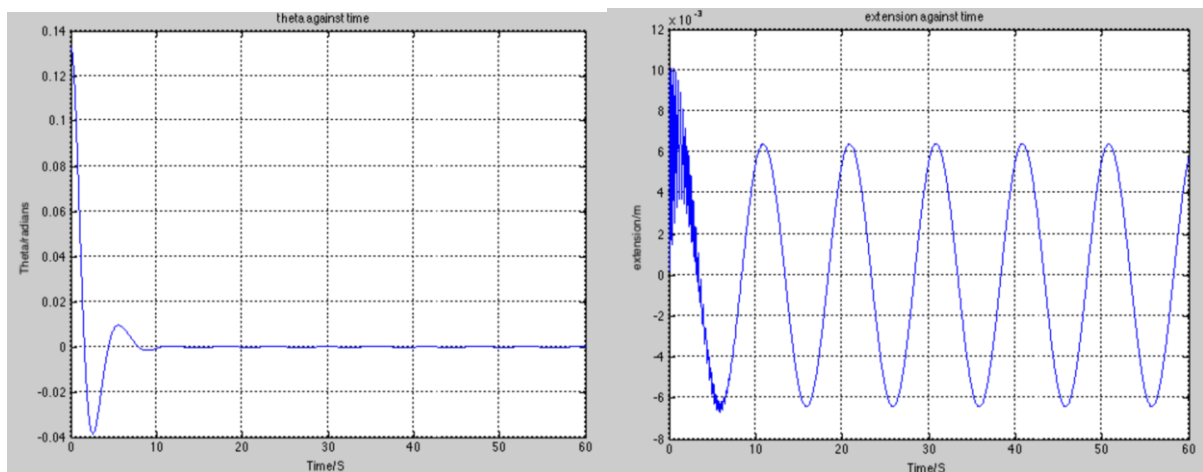
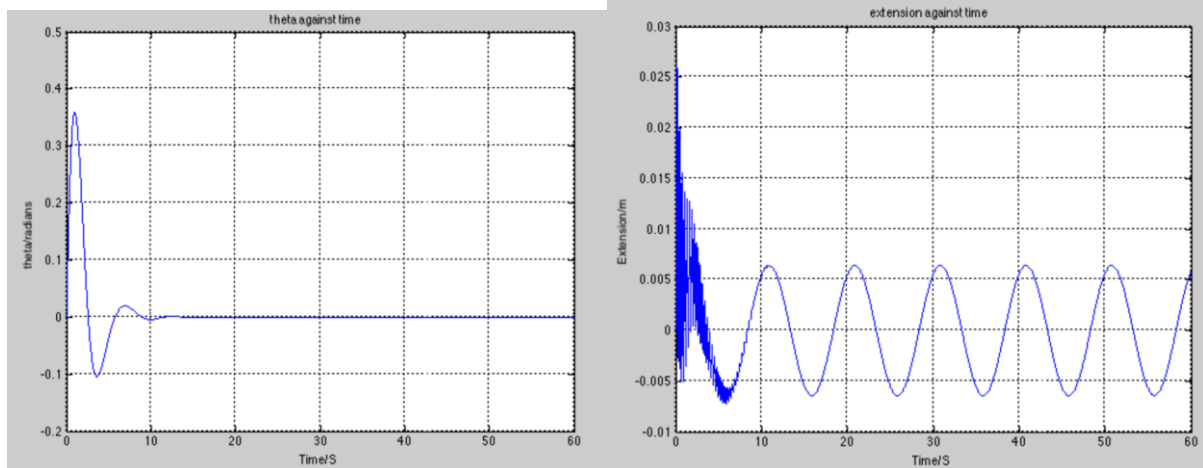


Figure 13, Solution of damped spring pendulum, subject to heave for initial conditions 2





For both solutions the steady state response of  $\theta$  is zero radians and is reached in 11s. The extension never reaches a steady state response. This is because once the energy from the initial conditions is dissipated overcoming the damping forces, energy is still provided from the vertical oscillation of the reference frame, once  $\theta$  has settled at zero this oscillation is solely in the radial component. Leading to a long-term response of a continuous periodic oscillation of period 10s and amplitude 6.4mm.

### 5.3. *Damped Spring Pendulum, Mother Ship Subject to Heave, with Winch Force*

This model was run with varying values for the fixed winch force, FW to properly assess the model. The results of which are shown in table 1. The solutions for a winch force of 2500N are shown in figures 14 and 15.

Figure 14, Solution of damped spring pendulum, subject to heave and winch force for initial conditions 1

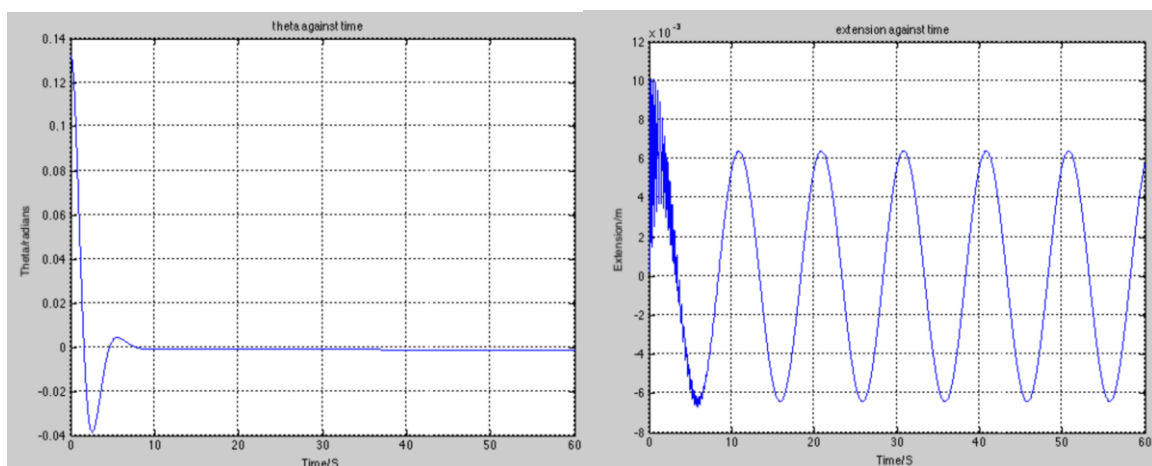


Figure 15, Solution of damped spring pendulum, subject to heave and winch force for initial conditions 1

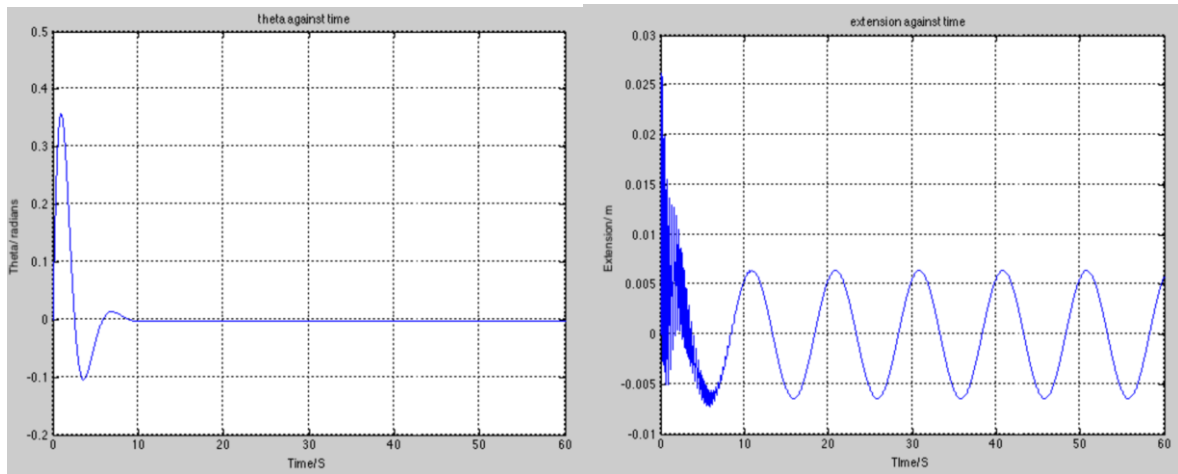


Table 1, Response of theta for varying winch forces

| Winch force/N | Value of theta at steady state response/rads | Time to steady state response/s |
|---------------|----------------------------------------------|---------------------------------|
| 500           | 0.0004                                       | 11                              |
| 700           | 0.0006                                       | 10                              |
| 1000          | 0.0009                                       | 8                               |
| 2500          | 0.0012                                       | 8                               |
| 5000          | 0.0015                                       | 6                               |

It was expected that the steady state response of theta to both sets of initial conditions would be zero radians. Table 1 show this not to be true suggesting that the numerical solution is unstable. The same correlation between winch force, the time to steady state response and its value was the same for solution 2. The solver used was explicit meaning as it tracks the Taylor series each calculation to a higher order relies on the previous calculation [22]. If the error grew with each step then the solution is unstable, which would cause the non-zero steady state response in theta calculated. The error increasing with an increasing winch force further backs this, assuming that the error term takes the same form as the solution [23]. The solution is stable as expected, however the computational scheme is unstable.

The winch force has little to no effect on the extension of the cable with the solutions mapping the same as those in section 5.2. This is because when resolving the winch force

only a small proportion of it is in the radial component of the system. Therefore the winch force in this component was too small to cause noticeable change in extension with such a large spring constant.

This instability in the solution was not a problem in the other models. Therefore it must be the added complexity of the winch leading to the instability. The steady state error in theta is small enough for the overall model to still be acceptable and have its uses.

#### 5.4. *Damped Spring Pendulum Mother Ship Subject to Pitch and Heave*

Figure 16, Solution of damped spring pendulum, subject to pitch and heave initial conditions 1

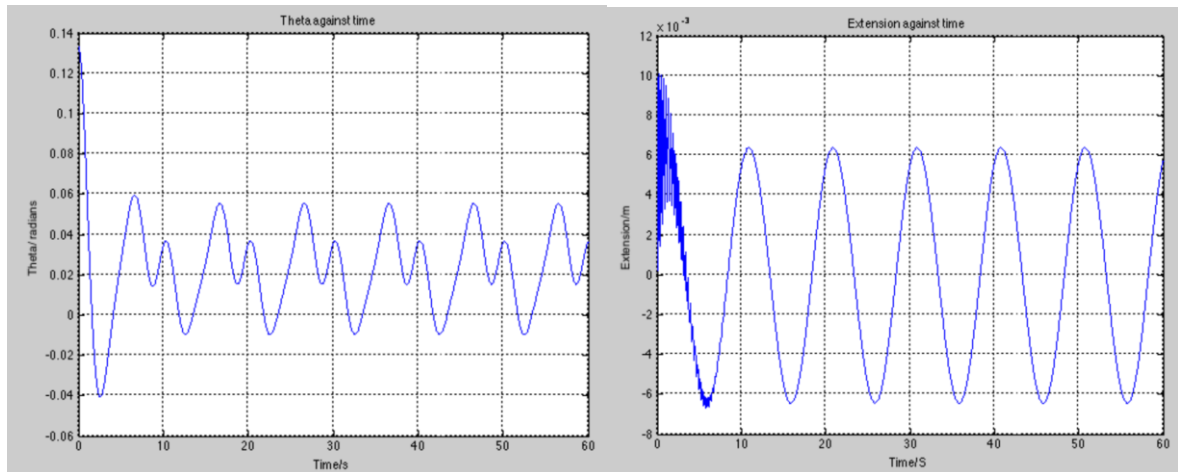
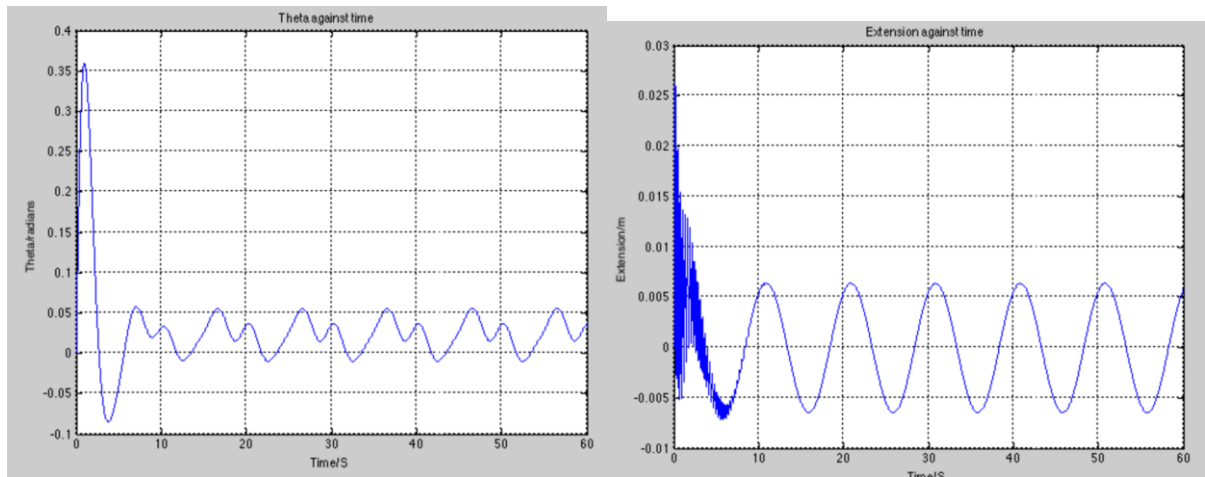


Figure 17, Solution of damped spring pendulum, subject to pitch and heave initial conditions 2



The response of the extension is the same as the response in section 5.2. The addition of a pitch function has lead to the response of theta no longer achieving a steady state response. This is because the rotation of the reference frame will constantly excite this output. The

double peak of  $\theta$ 's eventual periodic response is caused by the phase differences in the oscillation of the mass and that of the reference frame. The period of the oscillation of the long-term response of  $\theta$  is 10 seconds.

Comparing these results to those in section 5.2 it can be said that the response of extension is independent of the pitching function. This would be because the cable was modelled as a rigid spring so only forces in the radial component can affect the extension. Any affect of the pitching of the ship on forces in this component would not be noticed because of the relatively large spring constant. In reality the slackening of the cable would mean that the independence would not be the case.

### 5.5. *Damped Spring Pendulum, Mother Ship Subject to Pitch and Heave with Winch Force*

With the preliminary analysis conducted the final model could be run. For this model the winch force was set to 2500N to provide a noticeable change in the response of the degrees of freedom without too large an error.

Figure 18, Solution of damped spring pendulum, subject to pitch, heave and a winch force initial conditions 1

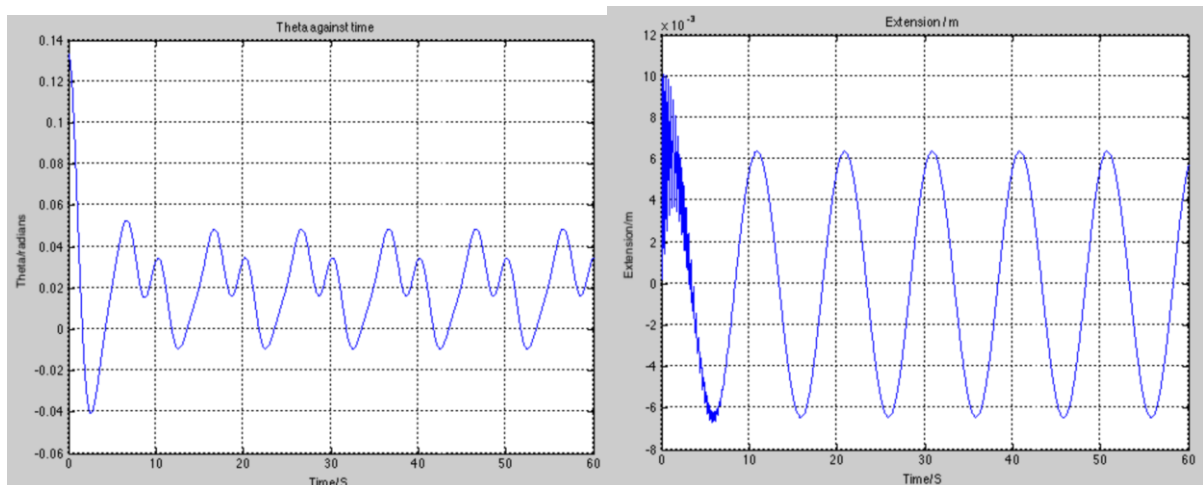
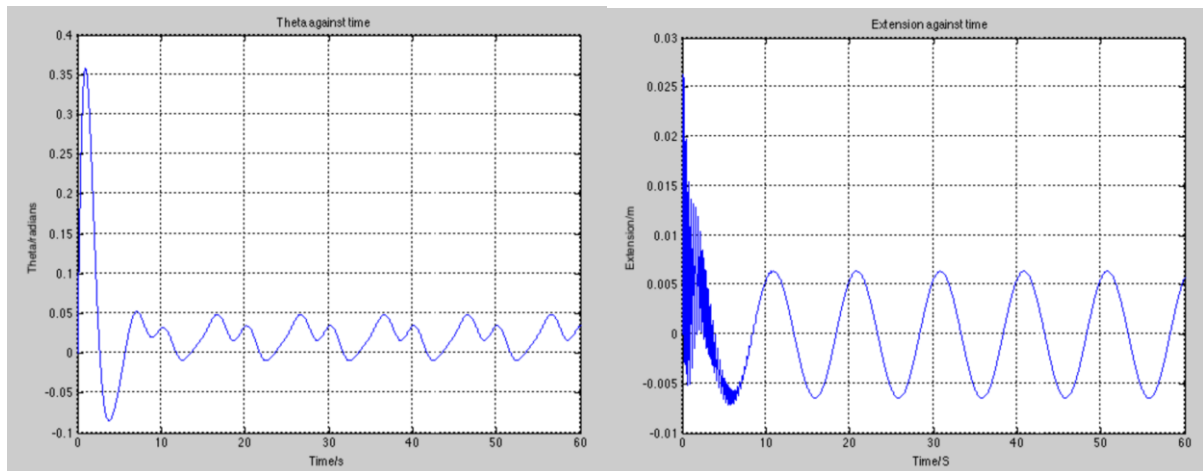


Figure 19, Solution of damped spring pendulum, subject to pitch, heave and a winch force initial conditions 2



The addition of the winch force has no effect on the response of the extension to the initial conditions. This would be because of the small proportion of force in this component and the relatively large spring constant as discussed in section 5.3.

There is a slight change in the response of theta with the maximum magnitude of the long-term response reducing by 0.005 radians and its period remaining unchanged.

The main importance to the operators of the NSRS is change of the resultant acceleration of the SRV as a result of the additional winch as it is this that affects the comfort of the rescued submariners and the difficulty in aligning the SRV with the transfer under pressure unit. These results are shown in figures 20 and 21.

Figure 20, Resultant acceleration against time for spring pendulum, pitch and heave initial conditions 1

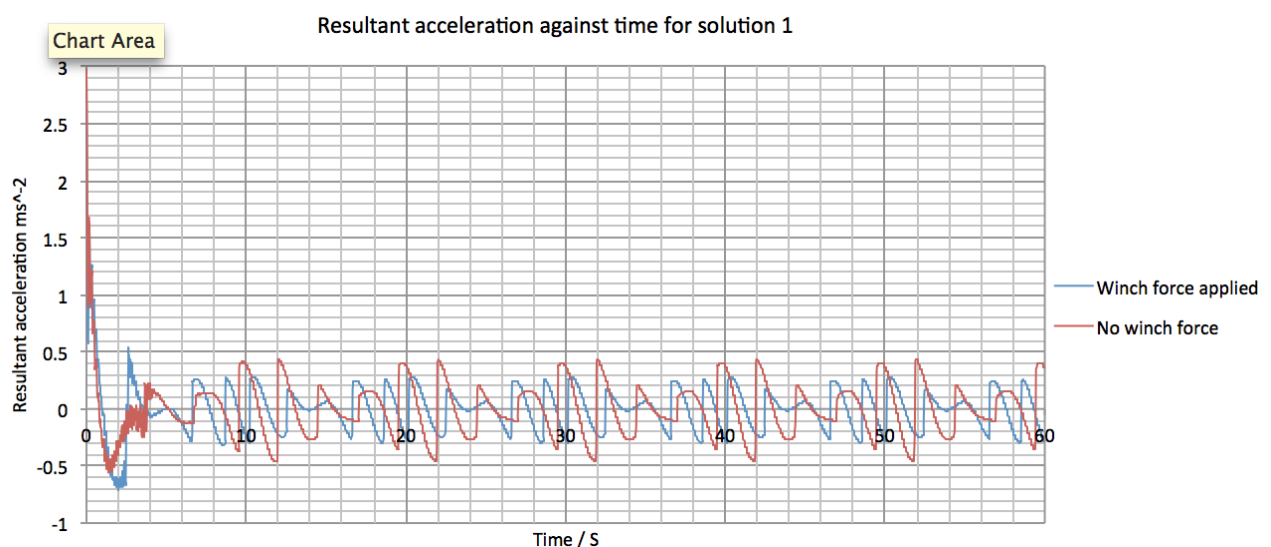


Figure 21, Resultant acceleration against time for spring pendulum, pitch and heave initial conditions 2

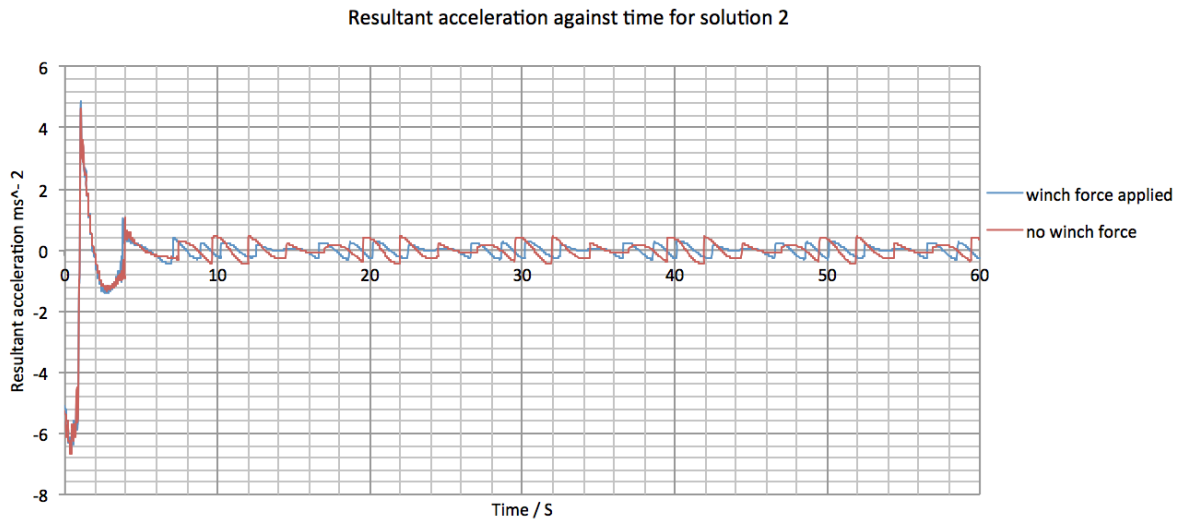


Table 2, maximum resultant acceleration magnitudes from figures 20 and 21

|                                         | Solution 1 |          | Solution 2 |          |
|-----------------------------------------|------------|----------|------------|----------|
|                                         | Winch      | No winch | Winch      | No winch |
| Short term response $ a_{max} /ms^{-2}$ | 2.78       | 2.97     | 6.56       | 6.68     |
| Long term response $ a_{max} /ms^{-2}$  | 0.295      | 0.451    | 0.295      | 0.451    |

Figures 20 and 21 show that the addition of the winch force for the most part for both sets of initial conditions results in a lower resultant acceleration of the SRV. The maximum magnitudes of the resultant acceleration for both the short term (transient) and long term responses reached for both solutions 1 and 2 were lower for the model including the winch force, shown in table 2. This is reasonable because of Newton's second law, the winch force counter acting the component of the weight swaying the SRV away from the ship will decrease the resultant force on the SRV, this in turn results in a lower acceleration as  $F=ma$ .

The much lower resultant accelerations for the long-term response of the resultant acceleration through the addition of a winch would be of benefit to the NSRS operators as it will make achieving the alignment of the SRV to the transfer under pressure unit [2] easier.

## 6. Discussion and conclusions

### 6.1. What has been learned

The 2-D equations of motion for a mass suspended on a spring with a force of fixed magnitude from a fixed origin in a reference frame that is subject to rotation and translation have been derived. This system was applied to model the launch and recovery of the SRV from a mother ship subject to pitch and heave, with an additional winch cable.

Due to their non-linearity the governing equations were solved numerically. Which has lead to finding system responses under two sets of initial conditions. When the system is damped and the movement of the reference frame is such to replicate the motion of the mother ship from the sea. Although the sea model was not as accurate as first intended the model used still provided a platform to assess the winch force model.

It was found that for a spring pendulum and a winch force of 2500N the effect of the winch force on the extension of the cable was negligible. And that if the mother ship is subject to pitch the extension is independent of the pitch function.

It was also found that if the mother ship is only subject to heave then a winch force of at least 700N is needed to reduce to the time to steady state response of theta to the initial conditions.

The main finding from the final model was that the additional winch force results in a lower resultant acceleration of the SRV for both sets of initial conditions.

## **6.2. *Potential for future research***

The NSRS can currently operate in up to a sea state 6 [1]. Operating in a time critical area increasing this limiting factor would be beneficial. This could potentially be achieved using an additional winch as modelled into this project. The governing equations derived in this project would provide a starting point for designing the necessary control algorithm alongside research being undertaken by SEA (Systems Engineering and Assessment), modelling the force a winch drum can create on a cable as a function of the torque on the winch drum.

Current developments in nonlinear control would need to be incorporated; feedback linearisation could potentially be used if system is deemed to be of controllable conical form [28] or flatness or sliding control [29]. Combining all of this could lead to a control system in which the tension on the winch cable is co ordinated in response to dynamics of the SRV. Potentially increasing the operational capabilities of the system. Success in this could be

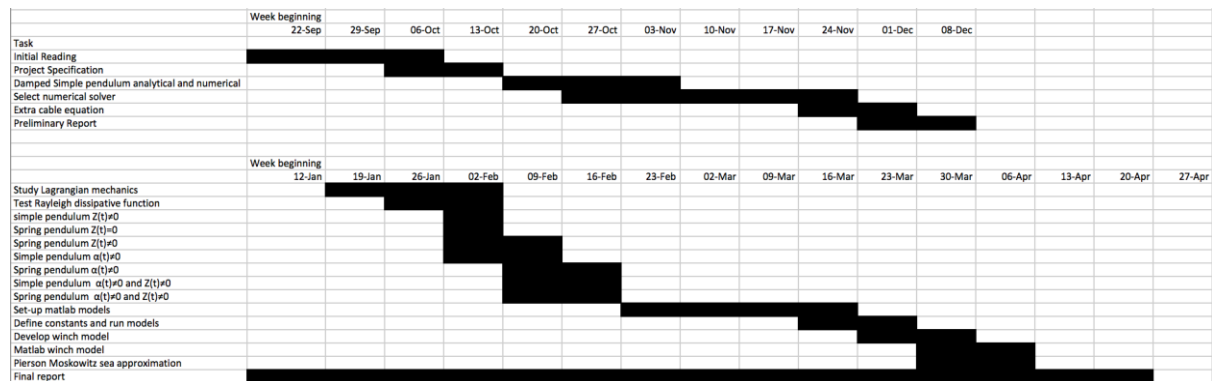
applied to other launch and recovery operations for the same benefits e.g. the sea fox mine disposal system.

## 7. Project management, consideration of sustainability and health and safety

### 7.1. Project management

The main source of project management was the use of a Gantt chart presented in figure 22. The purpose of the Gantt chart was to provide a visual reference of realistic time based targets to complete sections of the project. It evolved with the project as different sections of the project were altered and added or removed. For example it was only in the January that Lagrangian mechanics would be the best method for deriving the equations of motion for each model. Adhering to these alongside weekly supervisor meetings kept the project on track.

Figure 22, Gantt Chart



Another aspect of project management is to be aware of the potential risks to the completion of the project and to input precautionary measures. If the risk is unavoidable then a contingency plan is needed. These risks and measures are displayed in table 3.

Table 3, Potential risks to project

| Risk         | Probability | Impact   | Contingency/ Precaution                 |
|--------------|-------------|----------|-----------------------------------------|
| Illness      | Moderate    | Low      | Catch up with work when better          |
| Loss of work | Low         | Moderate | Save work regularly on multiple sources |



|                           |          |          |                                                          |
|---------------------------|----------|----------|----------------------------------------------------------|
| Inability to do a section | Moderate | Moderate | Adjust project accordingly so still merit in its outcome |
|---------------------------|----------|----------|----------------------------------------------------------|

## **7.2.      *Health and Safety***

As a modelling piece of work the majority of this project involved working with computers. The health implications of such can be found in Health and Safety (Display Screen Equipment) Regulations 1992 [24]. From this it can be said that when working with computers it is important to be in a comfortable position and take regular breaks.

When working in the computer rooms of Harrison it was important to be aware of the fastest exit from the occupied area. It was also important to know that in case of an evacuation the assembly point is the grass patch opposite the Harrison building. When working in the Harrison building out of hours the signing in and out process was followed [25].

With no self-imposed project specific risk being taken there was no need to conduct a risk assessment as all work was carried out within college procedures [26]. Which have been subject to risk assessments in align with the Health and Safety at Work Act 1974 [27].

## **7.3.      *Sustainability***

With nothing being manufactured sustainability was not a major concern for this project. However simple common sense energy saving measures were utilised. Turning the computer off if staying after the automatic computer shut down and turning off lights if the last one to leave a room.

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